# UNIT 2 APPROACHES TO SCIENTIFIC KNOWLEDGE: POSITIVISM AND POST POSITIVISM

**Structure**



## 2.0 OBJECTIVES

After going through this unit, you will be able to:

- Describe the aim of science as a cognitive enterprise;
- Explain the basic ideas of positivism;
- Appreciate the arguments or criticisms leveled against positivist philosophy of science;
- Explain the central tenets of Popper's philosophy of science;
- State the fundamental differences between Popper's and positivists' views about theory of scientific method;
- Assess Karl Popper's philosophy of science;
- Describe the essentials of Kuhn's philosophy; and
- Discuss the basic differences between the Popperian and the Kuhnian models of science.

## 2.1 INTRODUCTION

Science has not only shaped our mode of living in the world but also our ways of thinking about the world. It is for this reason it has acquired a central place in the intellectual life of our times. Because of its central place in the modern culture, three disciplines have emerged which has made science itself their object of inquiry. These three disciplines are: History of Science, Sociology of Science and Philosophy of Science. Whereas science studies the world natural or social or psychological, these three disciplines study science itself. History of Science and Sociology of Science are essentially twentieth century disciplines. No doubt, Philosophy of Science has a great past built by the contributions of individual philosophies in different centuries. However, as a widespread and coordinated discipline, it is also an essentially twentieth century phenomenon.

History of Science is a study of the development of scientific ideas and institutions in a chronological order. Sociology of Science is the study of the relation between social factors and forces on the one hand and the growth of scientific ideas and institutions, on the other. In short, it is an inquiry into the relation between science and society. History of Science deals with science as a historical phenomenon and sociology of science is concerned with science as a social institution. Philosophy of Science is a study of Science as a cognitive enterprise, that is, as a knowledge-seeking activity. The enquiry we call **Philosophy of Science** addresses questions like "What is the aim of Science?", "What is the method of Science?" In what sense, if any, is Science objective, rational and progressive?"

It is obvious that answers to questions such as these are of importance in figuring out how if at all, and in what sense, if any, social sciences including Economics, can claim to be scientific.

## 2.2 POSITIVIST PHILOSOPHY OF SCIENCE

Positivism (which is also called "logical positivism") is a movement in Philosophy of the first half of the 20th century. Positivists debunked the whole of traditional Philosophy by attacking Metaphysics which was the most important branch of Philosophy. Metaphysics was so central to Philosophy that Philosophy was virtually identified with Metaphysics. Metaphysics sought to answer most general and basic questions like "What is Ultimate Reality?" "Does God exist? And is so what is God's relation with the world?", "Is there a soul?", "What is the relation between the mind and the body?", "Is man free or are human actions determined?" etc., Positivists maintained that Metaphysics was a spurious discipline because metaphysical statements (such as "Matter is Ultimate reality" or "Ultimate Reality is Spiritual" etc.,) are meaningless since they are not verifiable in experience. "A statement" they claimed "is meaningful if and only if it is verifiable." Apart from being anti-metaphysical, they were empiricists i.e., according to them, experience is the source of knowledge. They called themselves "Neo-Empiricists" to distinguish themselves from the Traditional Empiricists of 17th and 18th centuries like Locke, Berkeley and Hume.

### 2.2.1 Central Tenets of Positivist Philosophy

In science we start with particular observations regarding what is the case. Using the method of induction, we arrive at definitions which are statements about the essential nature of things. We, then, using the method of deduction, on the basis of definitions, arrive at demonstrations which show why things must be what they are. Thus, the aim of science is two fold: **Definition and Demonstration**. The method of science, broadly, is of two types: **Induction and Deduction**. All science must proceed from "What is" given to us in particular observations to "what must be" as shown by the demonstrations. The path of science is an arch whose two end points are observation and demonstration.

In the whole span of three centuries – from the beginning of the 17th century till the end of the 19th century – two views stand out prominently as answer to the question regarding the aim and method of science. The first view is called Induction. The second view is called **Hypothesisism,** according to which the method of science is the method of Hypothesis. **Francis Bacon** is the Father of Inductivism and **Rene Dercartes** is the Father of Hypothesisism. The two views provide two distinct models of scientific practice, which we may call **Baconian** and **Cartesian** models of Scientific practice.

Positivists worked out a well-knit philosophy of science. Here are some of the central tenets of the Positivist Philosophy of Science:

1) Science is qualitatively distinct from, superior to and ideal for all other areas of human endeavor (**Scientism).**

2) The distinction, superiority and idealhood that Science enjoys is traceable to its possession of method (**Methodologism**).

3) There is only one method common to all sciences, irrespective of their subject matter (**Methodological monism**).

4) That method which is common to all sciences, natural or human, is the method of Induction (**Inductivism**).

5) The hallmark of science consists in the fact that its statements are systematically verifiable.

6) Scientific observations are or can be shown to be "pure" in the sense that they are theory-free.

7) The theories are winnowed from facts or observations.

8) The relation between observation and theory is unilateral in the sense theories are dependent on observations whereas observations are theory-independent.

9) To a given set of observation based statements, there corresponds uniquely only one theory (just as from a given set of premises in an argument, only one conclusion follows).

10) Our factual judgments are value-neutral and our value judgments have no factual content (Fact-Value Dichotomy thesis); hence, science being the foremost instance of factual inquiry, does not have value commitments.

11) That all scientific explanation must have the following pattern:

$L_1 \ldots\ldots\ldots\ldots L_n$

$I_2 \ldots\ldots\ldots\ldots I_n$

**Therefore, E.**

Where $L_1 \ldots\ldots\ldots\ldots L_n$ is a set of laws, $I_2 \ldots\ldots\ldots\ldots I_n$ is a set of statements describing initial conditions and E is the statement describing phenomenon to be explained. That is to say, to explain a phenomenon scientifically is to deduce its description from a set of laws (which are called "Covering Laws") via a set of statements describing initial conditions. In sum, all explanation worthy to be called '**scientific**' must contain laws and involve deduction (Hence, this is called Deductive-Nomologism where 'nomological' means 'concerning laws').

12) The aim of science is either economical description of phenomena or precise prediction of facts and not providing an account of observations in terms of unobservables. Hence, scientific theories are not putative descriptions of the unobservable world. The aim of science has nothing to do with alleged reality of such a world (Anti-Realism).

13) Unlike other areas of activity, science is progressive in the sense that scientific change is always change for better, whereas other areas exhibit just change: the progress of science consists in the accumulation of observations on the one hand, and, cumulative growth of theories,on the other hand. The latter means that any new theory includes the old theory(plus something). Thus the growth of science essentially exhibits continuity.

14) Science is objective in the sense that its theories are based on 'pure' observations or facts which as theory free. Interpretations may be subjective but observations/facts are objective because they are free from interpretation/ theory.

15) Science is rational because the principle of Induction which is central to the method of science is rationally defensible, in spite of Hume's skepticism regarding its rational defensibility.

Positivists tried to justify the principle of Induction by invoking the concept of pure observation. According to them, theories are arrived at on the basis of the Principle of Induction. If we can show that theories are very closely related to pure observations, the Principle of Induction stands rationally justified. They tried to work out a whole project to demonstrate rational justification of the Principle of Induction on these lines.

## 2.3 ATTACK ON THE POSITIVIST PHILOSOPHY OF SCIENCE

It must be noted that the concept of pure observation is necessary for positivist philosophy of science for showing that science is objective and that science is rational. Hence, the centrality of this concept to the positivist philosophy of science. Therefore, the collapse of this concept to the positivist philosophy of science though other theses of positivist philosophy of science listed above also were demolished by its opponents. Let us briefly look at the arguments which demolished the positivist thesis of pure observations i.e., thesis at Sl.No.6 in the list above.

**Firstly**, observations presuppose some principle of selection. We need relevant observations. In science it is the problem that decides what is a relevant observation and thus provides the principle of relevance. Hence, there can not be observations without a – prior problem.

As Popper says "Before we can collect data, our interest in data of a certain kind must be aroused; the problem always comes first". It may be objected that we become aware of the problems because of observations and hence observations come first and therefore positivists are right. But this objection does not hold. Two persons might make same observations but one may come out with a problem and the other may not. Therefore, mere observations would not generate problems in science. Usually problems are generated when there is a clash between what we observe and what we expect. Of the two persons making the same observations, one comes out with a problem because he sees a conflict between what he observes and what he expects whereas the other observer may have not expectation which conflicts with what he observes. The former believes in a theory which produces certain expectations which conflict with his observations and hence he comes out with a problem. In other words, a – prior belief in a theory is necessary for a problem to be generated and a – prior awareness of the problem is necessary for making relevant observations. Thus, theory precedes observations.

**Secondly**, in science observations are taken into account only if they are described in a language that is currently used in a particular science. An observation, however, genuine, is no observation unless it is expressed in a recognized idiom. It is the theory which provides the idiom or language to be used to describe facts or observations. It is relevant to quote the words of Pierre Duhem, a distinguished physicist and philosopher:

"Enter a laboratory. The table crowded with an assortment of apparatus: an electric cell, silk covered copper wire, small cups of mercury, spools, a mirror mounted on an iron bar; the experimenter is inserting into small openings the metal ends of ebony-headed pins: the iron bar oscillates and the mirror attached to it throws a luminous band upon a celluloid scale: the forward backward motion of this spot enables the physicist to observe the minute oscillations of the iron bar. But ask him what he is doing. Will he answer "I am studying the oscillations of an iron bar which carries a mirror?" No, he will say that he is measuring the electrical resistance of the spools. If you are astonished, if you ask him what his words mean, what relation they have with the phenomenon he has been observing and which you have noted at the same time as he, he will answer that your question requires a long explanation and that you should take a course in electricity".

**Thirdly**, most of the observations in science are made with the help of instruments. These instruments are constructed or designed in accordance with the specifications provided by some theories. These theories, one may say, form the software of these instruments. Belief in the reliability of these instruments implies the acceptance of these theories which has gone into the making of these instruments. This observations presuppose prior acceptance of theories.

**Fourthly**, observations in science need to be legitimized i.e., ratified by a theory. An example makes the point clear. We all know that Galileo used some telescopic observations to support the Helio-centric theory against the geo-centric theory of Ptolemy and his followers. His opponents did not consider the telescopic observations adequate. Why did they not? No doubt, they had belief in the reliability of telescope; they had no problem in using telescope for terrestrial purposes i.e., making observations of earthly objects. They opposed the extension of telescopic observations to the celestial sphere i.e., regarding heavenly bodies. Their argument was that the normal factors like background and neighbourhood which help our normal perceptions are absent in the sky. Further, it is impossible to directly verify whether telescopic observations of heavenly bodies are accurate. They rightly demanded from Galileo a theory of light which would justify the extension of the telescope from terrestrial to celestial sphere. Galileo had no such theory. But he rightly believed that such a theory could be provided in future so that telescopic observations would get justification. Thus, while the opponents of Galileo insisted that the telescopic observations be justified by an optical theory at the same time as their acceptance, Galileo maintained that the justification could be provided subsequent to their acceptance. It may be noted that both sides accepted that the telescopic observations needed justification in term of a theory of light.

All this does not mean that observations are theory-dependent whereas theories are observation-independent. Observations and theory are interdependent though it is not easy to clarify what the nature of this interdependence is. However, positivists were wrong in claiming that observations are theory –independent. To say that observations are theory dependent is to say that observation is not a passive reception but an active participation of our cognitive faculties equipped with prior knowledge which we call theory. After all, observations are not 'given' but 'made'.

**Check Your Progress 1**

1) What is the aim of science?

......................................................................................................................

......................................................................................................................

......................................................................................................................

......................................................................................................................

......................................................................................................................

2) What do you mean by the term 'inductivism'?

......................................................................................................................

......................................................................................................................

......................................................................................................................

......................................................................................................................

3) What are the features of hypothesis?

......................................................................................................................

......................................................................................................................

......................................................................................................................

......................................................................................................................

4) Match the following:

| A | B |
|---|---|
| i) The relation between observation and theory | i) Value judgments |
| ii) Factual inquiry under positivist approach does not have | ii) Precise prediction of facts |
| iii) The aim of Science is | iii) Objective |
| iv) Science is | iv) Unilateral |

5) Give two counter arguments against the positivist thesis of pure observation.

.....................................................................................................................

.....................................................................................................................

.....................................................................................................................

## 2.4 KARL POPPER'S PHILOSOPHY OF SCIENCE

Karl Popper was the first to react against the positivist philosophy of science. In fact he started attacking it quite early. But his attack on positivists and his alternative to the positivist philosophy of science came to be widely known at the beginning of the second half of the 20th century. Popper's theory of science, particularly his theory of scientific method has won a lot of admirers among scientists and philosophers. As we know, positivists tried to work out a **sophisticated version of Inductivism**. Popper worked out a **sophisticated version of Hypothesisism**. We shall briefly consider his views on the nature of science.

According to Popper, the central task of philosophy is not to solve Hume's problem or problem of Induction as thought by Positivists. This is because (1) the problem of Induction cannot be solved, and (2) it need not be solved because the method of science is not the method of Induction. The central task of philosophy of science, Popper maintains, is to solve what he calls the problem of demarcation or Kant's problem i.e., the problem of identifying the line of demarcation between science and non-science. Popper maintains that **what distinguishes science from the rest of our knowledge is the systematic falsifiability of scientific theories**. This falsifiability is the line of demarcation between science and non-science. Falsifiability is the criterion of scientificity. A statement is scientific if and only it is falsifiable.

In accordance with what he considers to be the hallmark of scientific theories, Popper puts forward what he considers to an adequate model of scientific method. He characterizes his model of scientific method as **Hypothetico-Deductive model**. According to him, the method of science is not method of Induction but the method of **Hypothetico-Deduction**. What are the fundamental differences between these methodological models"? Firstly, **the inductivist model maintains that our observations are theory-independent and therefore are indubitable. That is to say, since observations are theory-independent, they have probability Value 1**. It also says that our theories are only winnowed from observations and therefore our scientific theories have the initial probability value 1 in principle. Of course, inductivists admitted that in actual practice, the theories may contain something more than what observation based statements say, with the result that our actual theories may not have been winnowed from observations.

Hence, the need for verification arises. Popper rejects the inductivist view that our observations are theory-free and hence rejects the idea that our observation statements have probability equal to 1. More importantly, he maintains that theories are not winnowed from observations or facts, but are free creations of human mind. Our scientific ideas, in other words, are not extracted from our observations; they are pure inventions. Since **our theories are our own**

**constructions, not the functions of anything like pure observations, which according to Popper are anyway myths, the initial probability of our scientific theories is zero**.

From this it follows that whereas according to the inductivists what scientific tests do is to mrely find out whether our scientific theories are true, according to Popper scientific tests cannot establish the truth of scientific theories even when the tests give positive results. If a test gives a positive result, the inductivists claim that the scientific theory is established as true, whereas according to Popper all that we calim is that our theory has not yet been falsified. In Popper's scheme no amount of positive result of scientific testing can prove our theories. Whereas the inductivists speak of confirmation of our theories in the face of positive results of the tests, Popper only speaks of corroboration. In other words, in the inductivist scheme we can speak of scientific theories as established truths, whereas in the Popperian scheme, a scientific theory, however, well supported by evidence remains permanently tentative. We can bring out the fundamental difference between verificationism (inductivism) and falsificationism (Hypothetico-Deductivism) by drawing on the analogy between two systems of criminal law. According to one system, the judge has to start with the assumption that the accused is innocent and consequently unless one finds evidence against him, he should be declared innocent. According to the other, the judge has to start with the assumption that the accused is a culprit and consequently, unless evidence goes in his favour, he should be declared to be a culprit. Obviously the latter system of criminal law is harsher than the former. The inductivist scheme is analogous to the former kind of criminal law, where as the Hypothetico-Deductive scheme is akin to the latter one.

**The steps of Scientific Procedure**

In the Popperian scheme we begin with a problem, suggest a hypothesis as a tentative solution, try to falsify our solution by deducing the test implications of our solution, try to show that the implications are not borne out and consider our solution to be corroborated if repeated attempts to falsify it fail. Thus, problem, tentative solution, falsification and corroboration constitute the steps of scientific procedure. Popper's theory of scientific method is called Hypothetico-Deductivism because, according to him, the essence of scientific practice consists of deducing the test implications of our hypothesis and attempting to falsify the latter by showing that the former do not obtain, whereas according to Inductivism the essence of scientific practice consists of searching for instances supporting the generalization arrived on the basis of some observations and with the principle of induction.

Popper claims that the Hypothetico-Deductive model of scientific method is superior to inductivist model for the following reasons:

**Firstly,** it does justice to the critical spirit of science by maintaining that the aim of scientific testing is to falsify our theories and by maintaining that our scientific theories are, however corroborated, going to permanently remain tentative. In other words, the Hypothetico-Deductivist view presents scientific theories as permanently vulnerable with the sword of possible falsification always hanging on their head. The inductivist view of scientific method makes science a safe and defensive activity by portraying scientific testing as a search for confirming instances and by characterizing scientific theories as established truths. According to Popper, the special status accorded to science is due to the fact that science embodies an attitude which is essentially open-minded and anti-dogmatic. Hypothetico-Deductivism is an adequate model of scientific practice because it gives central place to such an attitude.

**Secondly,** Popper thinks that if science had followed the inductivist path; it would not have made the progress it has. Suppose a scientist has arrived at a generalization. If he follows the inductivist message, he will go in search of instances which establish it as a truth. If he finds an instance which conflicts with his generalization, what he does is to qualify his generalization saying that the generalization is true except in the cases where it has to be held unsupported. Such qualifications impose heavy restrictions on the scope of the generalization. This results in scientific theories becoming extremely narrow in their range of applicability. But if a scientist follows the Hypothetico-Dedectivist view, he will throw away his theory once he comes across a negative instance instead of pruning it and fitting it with the known positive facts. Instead of being satisfied with a theory, tailored to suit the supporting observations, he will look for an alternative which will encompass not only the observations which supported the old theory but also the observations which went against the old theory and more importantly which will yield fresh test implications. The theoretical progress science has made can be explained only by the fact that science seeks to come out with bolder and bolder explanations rather than taking recourse to the defensive method of reducing the scope of the theories to make them consistent with facts. Hence, Popper claims that the Hypothetico-Deductive model gives an adequate account of scientific progress. According to him, if one accepts the inductivists account of science one fails to give any explanation of scientific progress.

**Thirdly**, the **Hypothetico-Deductive** view according to Popper avoids the predicament encountered by inductivist theory in the face of **Hume's** challenge. As we have seen, Hume conclusively showed that the principal of induction could not be justified on logical grounds. If Hume is right, than science is based upon an irrational faith. According to Hypothetico-Deductivist view, science does not use the principle of induction at all. Hence, even though Hume is right, it does not matter since science follows the Hypothetico-Deductivism are so radically different that the latter in no way face any threat akin to the one faced by the former. In this connection, he draws our attention to the logical asymmetry between **verification**, the central component of the inductivist scheme, and **falsification**, the central component of the Hypothetico-Deductivist scheme. They are logically asymmetrical in the sense that one negative instance is sufficient for conclusively falsifying a theory, whereas no amount of positive instances are sufficient for conclusively falsifying a theory, whereas no amount of positive instances are sufficient to conclusively verify a theory. It may be recalled that Hume was able to come out with the problem of induction precisely because a **generalization** (all theories according to Inductivism are generalizations) cannot be conclusively verified.

How does Popper characterize scientific progress? According to him, one finds in the history of science invariable transitions from theories to better theories. What does the work 'better' stand for? It may be recalled that, according to Popper, no scientific theory, however, corroborated can be said to be 'true'. Hence, Popper drops the very concept of truth and replaces it by the concept of **Verisimilitude** (truth-likeness or truth-nearness) in his characterization of the goal of science. In other words, through science cannot attain truth, i.e., though our theories can never be said to be true, science can set for itself the goal of achieving higher and higher degrees of Verisimilitude, i.e., successive scientific theories can progressively approximate to truth. So, in science we go from theory to better theory and the criterion for betterness is Verisimilitude. But what is the criterion for Verisimilitude? The totality of the testable implications of a hypothesis constitutes what he calls 'the empirical content' of the hypothesis. The totality of the testable implications which are borne out constitute the truth content of the hypothesis and the totality of the testable implications which are not borne out is called the falsity content of the hypothesis. The criterion of the Verisimilitude of a theory is nothing but truth content minus the falsity content of a theory. In the actual history of science we always find, according to Popper, theories being replaced by better theories, that is, theories with higher degree of Verisimilitude. In other words, of the two successive theories, at any time in the History of Science, we find the successor theory possessing greater Verisimilitude and is therefore better than its predecessor. In fact, according to him, **a theory is rejected as false only if we have an alternative which is better than the one at hand** in the sense that it has more testable implications and a greater number of its testable implications are already borne out. The growth of science is convergent in the sense that the successful part of the old theory is retained in the successor theory with the result the old theory becomes a limiting case of the new one. The growth of science thus shows continuity. In other words, it is the convergence of the old theory into the new one that provides continuity in the growth of science. It must also be noted in this connection that unlike the Inductivists or Positivists, Popper is a **Realist** in the sense, according to him, scientific theories are about an unobservable world. This means that the real world of the unobservable thought can never be captured entirely by our theories, evidence that through the gap between Truth and our theories can never be completely filled, it can be progressively reduced. Consequently, the real world of unobservable will be more and more like what our theories say though not completely so.

How does Popper establish the objectivity of scientific knowledge? Inductivists sought to establish the objectivity of science by showing that scientific theories are based upon pure observations. The so-called 'pure' observations were supposed to be absolutely theory-free. They are only 'given' and hence, free from subjective interpretations. Popper, as we have seen, rightly rejects the idea of pure observations. Consequently, he cannot accept the inductivist account of the objectivity of science. What engenders scientific objectivity, according Popper, is not the possibility of pure observation, but the possibility of **inter-subjective testing**. In short, science is objective because it is public and it is public because its claims are inter-subjectively testable.

To the question, "Which comes first, observation or theory?" the Inductivist answers 'observation'. Popper answers 'earlier observation or earlier theory'. To him the question is as illegitimate as the question, "which comes first, egg or hen?" which can be answered only by saying 'earlier egg or earlier hen?'

It will be convenient if we list the main theses of Popper's philosophy of science arranged in a manner with our list of the theses of the positivist philosophy of science: